History & Document Approval

Version	Date	Author	Description
1.0	18-Oct-2026	Muhammad Aiman Hafiz Bin Rosli	Initial Draft - Chapters 1, 2 & 3 (3.1-3.3) for UpsiRent-I

1.0 Introduction

UpsiRent-I is a digital platform designed for accommodation searching for students in helping to increase efficiency, safety and critical decision-making. This web contains property listings, roommate advertisements and calculators that make house hunting into enjoyable experiences. This web is also used by students as an interactive rental and assessment tool that makes finding accommodations secure and engaging.

1.1 Purpose

The purpose of the UpsiRent-I is to:

- To make student house hunting safe, engaging and effective by combining accommodation searching with smart tools.
- Turn the traditional rental searching into enjoyable experiences.
- Help students understand rental contracts more easily and encourage active scam-prevention instead of passive renting.
- Allow students to search for properties anytime and anywhere at their own pace, supporting independent and flexible planning.

1.2 Scope of the Project

The scope of the UpsiRent-I is:

- Students and external guests can utilize the AI assistant to identify potential red flags and suspicious clauses in rental agreements to avoid scams.
- Users can protect themselves from fraudulent activities by studying the provided anti-scam guides and verified landlord checklists.
- Individuals can prevent hidden fees and ensure transparent utility sharing by calculating accurate rent estimates before committing to a tenancy.

1.3 Definitions, Acronym and Abbreviations

Term	Definition
Gemini AI	Gemini AI (or artificial intelligence) is a concept that combines advisory and automation to make rental screening more engaging and safe.
UpsiRent-I	UpsiRent-I is an accommodation website that uses interactive and smart elements to make searching Malaysian student rentals entertaining, interesting, and easier for students, helping them secure better housing.

1.4 References

1. index.html – The front-end user interface prototype for UpsiRent-I.
2. Personal Data Protection Act 2010 (PDPA) / Akta 709 (Malaysia).
3. Google Gemini API Documentation.
4. IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018 – Standard for Software Requirements Specifications.

1.5 Document Overview

This document presents a detailed overview of the UpsiRent-I system, detailing its core objectives, primary functionality, and target user groups. Section 1 establishes the system's foundation by outlining its purpose, scope, and key terminology to ensure a clear context for readers.

The remaining sections explore the platform's architectural structure and technical design in greater depth:

- Section 2 provides a general product description, examining product perspectives, system capabilities, target user profiles, and operational constraints or dependencies.
- Section 3 details specific requirements, including external system interfaces, functional capabilities, use cases, technical specifications, and system diagrams.

The document concludes by addressing quality assurance measures to guarantee that all specified requirements directly support the project's overall goals. Together, these sections establish the UpsiRent-I SRS as an essential guide for stakeholders during development, testing, and implementation.

CHAPTER 2: General Description

This chapter offers an overview of the UpsiRent-I web application, detailing its operating context and user interactions. It outlines how student users search for rental accommodations, homestays, and roommates. Additionally, it defines key stakeholders, user categories and characteristics, and the primary functional capabilities of the system.

2.1 Product Perspective

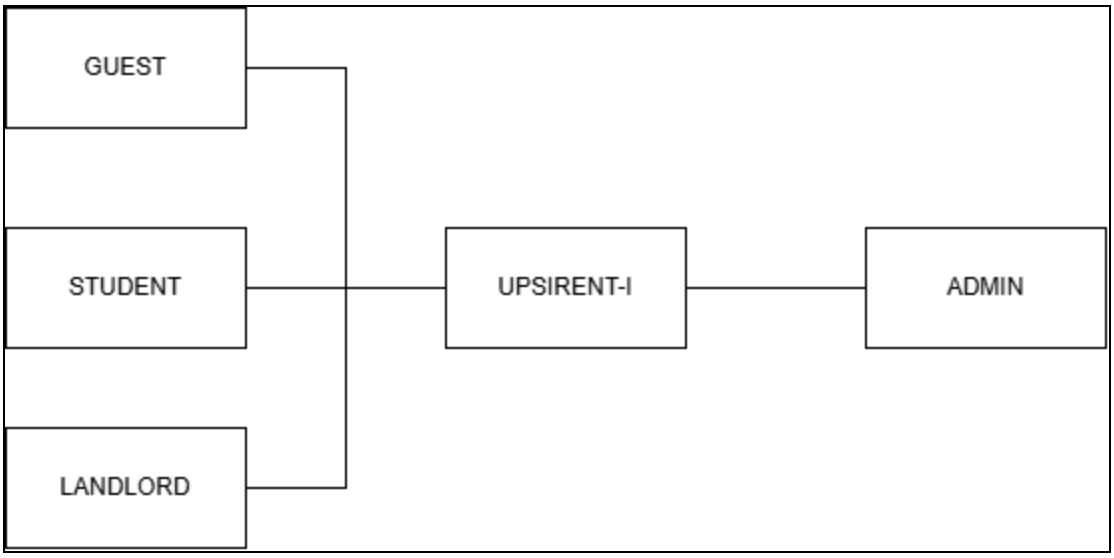


Figure 2.1: Product Perspective for UpsiRent-I

The UpsiRent-I is an interactive accommodation portal designed to help university students find suitable rental houses, homestays, and roommates. Students can easily explore available properties using features such as search filters, room photo galleries, and a rent calculator. These features make it easier for students to compare their options and choose accommodation that suits their needs and budget. Meanwhile, the administrator manages the platform by verifying student IDs, reviewing and approving or rejecting property advertisements, and managing user accounts. These tasks help ensure that the platform remains safe, reliable, and up to date while meeting the accommodation needs of the university community.

2.2 Product Functions Description

2.2.1 User Function

2.2.1.1 User Authentication and Profile

This function provides a secure login and registration system for students and landlords to access the UpsiRent-I platform based on their respective roles. It ensures that user data, wishlists, and submitted advertisements are securely linked to individual profiles.

2.2.1.2 Property Search and Filtering

This function allows users to browse "Rumah Sewa", "Homestay", and "Cari Rakan Bilik" listings. Users can easily filter the properties by location, availability status, maximum budget, and keyword search to find accommodations that match their specific needs.

2.2.1.3 View Property Details and Maps

This function allows users to view detailed property information in an interactive modal. It showcases a room-by-room photo gallery, a complete inventory checklist, landlord details, tenant reviews, and provides quick links to external Google Maps for location accuracy.

2.2.1.4 Student Verification

This function allows students to submit a verification form by uploading their UPSI Student ID or Offer Letter. The uploaded documents will be sent to the administrator for manual verification to unlock a special 10% discount on all rental prices.

2.2.1.5 Gemini AI Assistant

This function provides users with an interactive AI chat assistant to help them screen rental contracts for red flags, estimate monthly living costs, and find suitable housing. It utilizes a backend Gemini AI proxy with a local fallback engine if the API is unavailable.

2.2.1.6 Rent Calculator

This function provides an interactive calculator for users to estimate the per-head rental cost. It allows users to adjust sliders for base rent and capacity, tick checkboxes for shared utilities, and generate a summary to share directly to WhatsApp.

2.2.1.7 Property Comparison

This function allows users to select a minimum of two and a maximum of three properties to view them side-by-side. It displays a comparison modal covering price differences, distance to campus, capacity, and contract terms.

2.2.1.8 Wishlist Management

This function allows registered students to save their favorite properties to a personalized wishlist. The saved properties are stored persistently in the backend database for easy future access.

2.2.1.9 Landlord Property Submission

This function allows landlords to submit and manage property advertisements. Landlords can fill in property details and upload property photos, which will then undergo manual approval by the administrator before being published to the public.

2.2.1.10 Roommate Hub

This function allows registered students to view and publish roommate advertisements on a community board. All submitted advertisements will be saved to the database and require administrator moderation to maintain a safe environment.

2.2.1.11 View Info Guide

This function provides users with interactive modals containing essential student guides. The guides include UPSI Free Shuttle Bus routes, anti-scam tips for renting, and information on nearby public amenities.

2.2.2 Admin Function

2.2.2.1 Admin Moderation and Management

This function allows administrators to manage the overall platform by reviewing and approving or rejecting student verification documents, property advertisements, and roommate ads. It also allows the admin to manage, suspend, or delete user accounts to ensure system security.

2.3 User Characteristics

2.3.1 User Classes Definition

User Class	Description
Guest	Unregistered users or outsiders. Can browse listings, view property details, and use the calculator. Cannot access the 10% student discount or save properties.
Registered Student	UPSI or IPT students who have successfully registered an account and logged in. They can save properties to the Wishlist ("Disimpan") and post roommate advertisements ("Cari Rakan Bilik").
Verified Student	Registered students who have submitted their Student ID / Offer Letter and received manual approval from the Admin. They gain access to the **10% discounted pricing** automatically applied across all listings.
Landlord	Property owners or agents. They can submit property listings, upload photos, and manage their own listings upon approval.
Admin	System administrators created internally by the system. Responsible for manually verifying student IDs, moderating property/roommate advertisements, and managing user accounts.

2.3.2 User Access Functions

Function	Guest	Student	Landlord	Admin
Register / Login	/	/	/	/
Search & Filter Properties	/	/	/	/
View Property Details & Maps	/	/	/	/
Rent Calculator	/	/	/	/
Property Comparison	/	/	/	/
Gemini AI Assistant	/	/		
View Info Guide	/	/	/	/
View Roommate Board	/	/	/	/
Wishlist Management		/		
Post Roommate Ads		/		
Submit			/	

Property Listing				
Student Verification Form		/		
Approve Ads & Verification				/
Manage Users				/

2.4 Constraints, Assumptions and Dependencies

CR_ID	Constraint Requirement
CR001	Internet connectivity is required for all system functionalities, particularly the AI proxy and map routing.
CR002	The system must be fully responsive and compatible with modern web browsers on desktop and mobile devices.
CR003	**Security:** API keys and sensitive environment variables must be securely handled on the backend server, strictly prohibiting exposure on the client-side.
CR004	**PDPA Compliance:** The collection of Student IDs, Landlord documents, and the display of user WhatsApp numbers must strictly comply with PDPA 2010. Explicit consent must be obtained before exposing

	personal contact details.
CR005	**Content Moderation:** All submitted property listings and roommate advertisements must undergo manual moderation by an Admin before publication to prevent scams and inappropriate content.

Assumptions and Dependencies:

1. **WhatsApp Dependency:** The system assumes users have WhatsApp installed on their devices, as the primary communication method between tenants, landlords, and roommates relies on the wa.me URL scheme.

2. **API Availability:** The AI features depend on the continued availability and stability of the Google Gemini API.

3. **Manual Moderation:** The system assumes that System Admins will actively log in to manually review and approve student verifications, property listings, and roommate advertisements within a reasonable timeframe.

CHAPTER 3: Specific Requirements

3.1 External Interface Requirements

3.1.1 System Interfaces

UpsiRent-I is a web-based application accessible via modern web browsers including Google Chrome, Safari, Firefox, and Edge. The user interface is designed with Tailwind CSS to ensure a fully responsive experience across desktop and mobile devices. The system shall interface securely with a backend server and a relational database (such as MySQL) for persistent data storage, user management, and seamless real-time data retrieval.

3.1.2 Hardware Interfaces

There are no specific hardware interfaces required other than standard internet-enabled devices:

- **Desktop/Laptop:** Minimum 2GB RAM, modern web browser.
- **Mobile Devices:** Android or iOS devices with modern mobile browsers.

3.1.3 Communication Interfaces

- **HTTPS:** All data transmitted between the client browser and the server (especially login credentials and Student ID uploads) must be encrypted using HTTPS.
- **WhatsApp URL Scheme:** The system uses https://wa.me/ links to pass pre-filled message templates from the application directly to the user's WhatsApp application.
- **Google Maps URL Scheme:** The system redirects users to https://www.google.com/maps/search/ using specific query parameters for location viewing.

3.1.4 Software Interfaces

- **Gemini AI Backend Proxy:** The system backend shall communicate with the Google Gemini API to generate responses. The API calls must be executed strictly from the server-side to protect API keys. The AI model version must be configurable in the backend environment settings (not hardcoded) to allow for future model upgrades. The backend will inject the latest property data from the database into the system context before sending the prompt to the AI.

3.2 User Interfaces

3.2.1 Screen Flow Descriptions

3.3 Functional Requirements

FR_ID	Description
FR001	Authentication & Profile
FR002	Role-Based Access Control (RBAC)
FR003	View & Switch Listing Categories
FR004	Search, Filter & Reset
FR005	Property Details & Maps

FR006	Tenant Reviews
FR007	Rent Calculator
FR008	Property Comparison
FR009	Student Verification Submission
FR010	Discount Application
FR011	Gemini AI Chat
FR012	Wishlist
FR013	Submit Property Listing & Photos
FR014	Host Document Verification
FR015	Manage Own Listings
FR016	Roommate Hub
FR017	Submission Status Tracking
FR018	In-App Notifications
FR019	PDPA Consent Check
FR020	Data Retention & Deletion
FR021	Admin Moderation Dashboard
FR022	Manage Users
FR023	Large Text Mode (Accessibility)
FR024	Info Guide

3.4: Use Cases

3.4.1 Use Case Diagrams (Main)

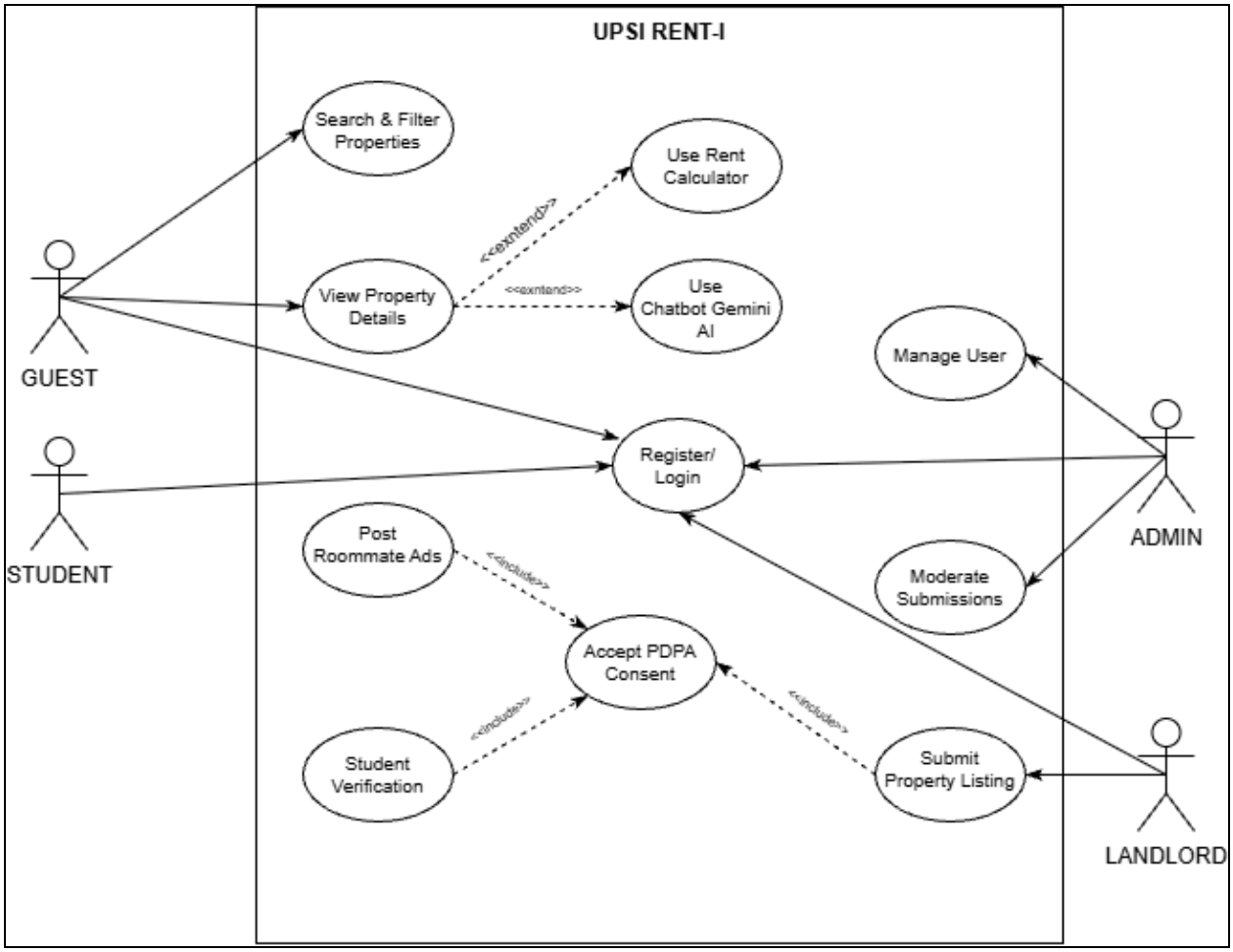


Figure 3.4.1: UpsiRent-I Use Case Diagram (Main)

The use case diagram identifies four primary actors within the system. The Guest, an unauthenticated visitor, may register an account, search and filter property listings, and view detailed property information. The Student, an authenticated user, inherits all Guest capabilities and may additionally submit a student verification request and publish roommate advertisements. The Landlord is responsible for submitting property listings to the platform. The Administrator, a privileged user, moderates all user-generated submissions, including student verifications, property listings, and roommate advertisements, and also manages user accounts. All form submissions are subject to a mandatory PDPA consent check, ensuring compliance with personal data protection requirements before any data is collected. In addition, the Rent Calculator and the Gemini AI assistant are available as optional tools when viewing a property,

allowing users to access supplementary functionality as needed without affecting the core flow.

3.4.1.1 Authentication Subsystem Diagram

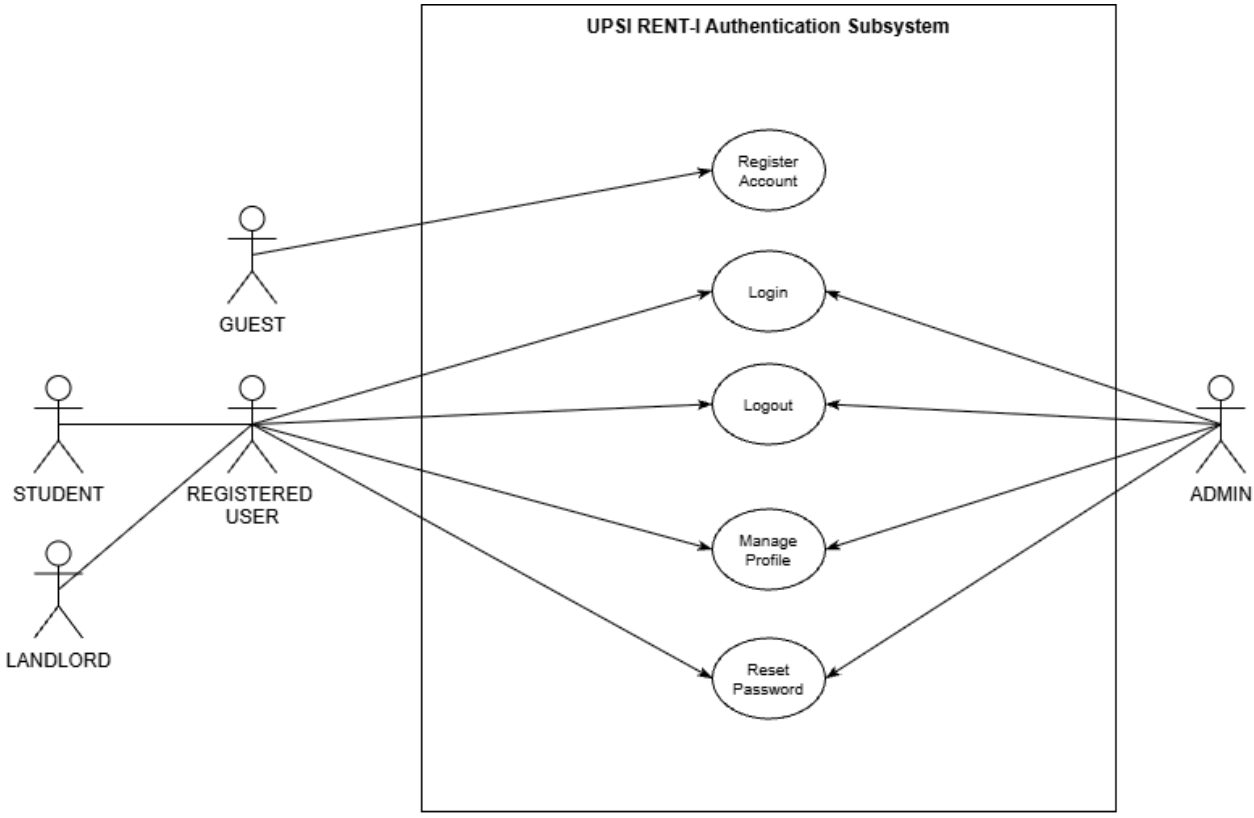


Figure 3.4.1.1: UpsiRent-I Use Case Diagram for Authentication

3.4.1.2 Property & Rental Subsystem

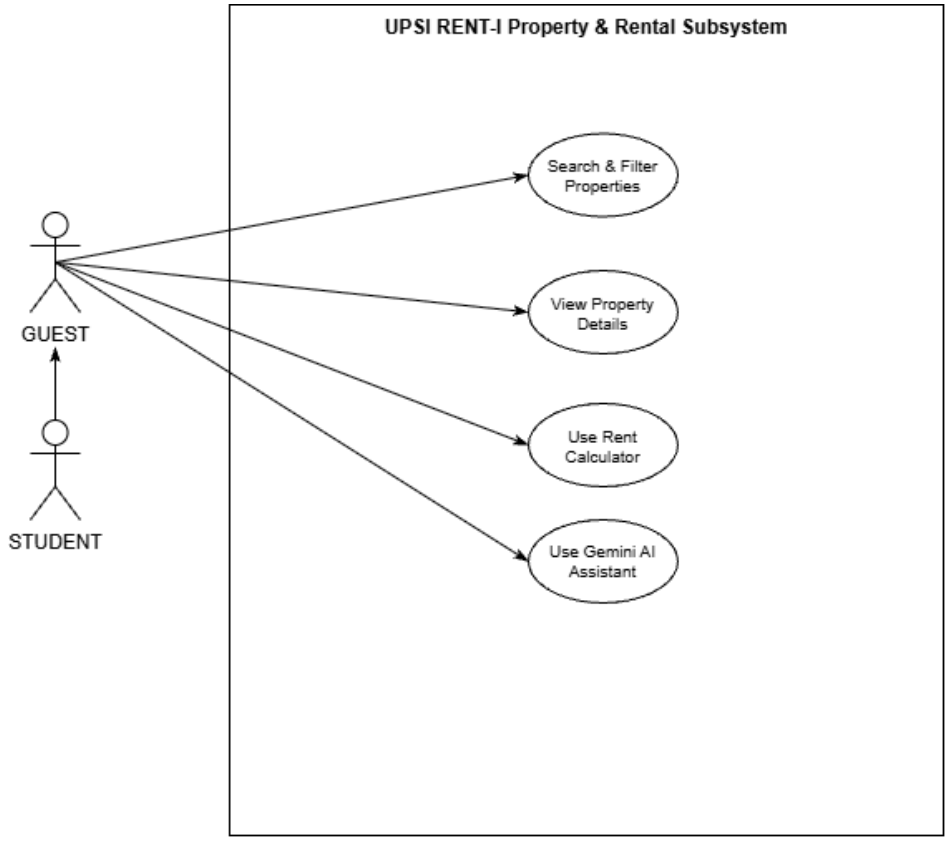


Figure 3.4.1.2 Property & Rental Subsystem

3.4.1.3 Landlord Listing Subsystem

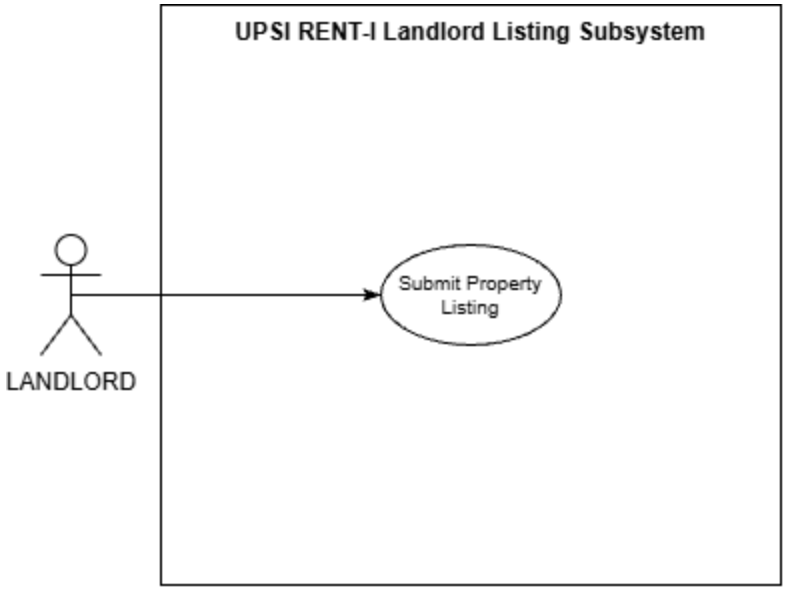


Figure 3.4.1.3 Landlord Listing Subsystem

3.4.1.4 Student Community & Verification Subsystem

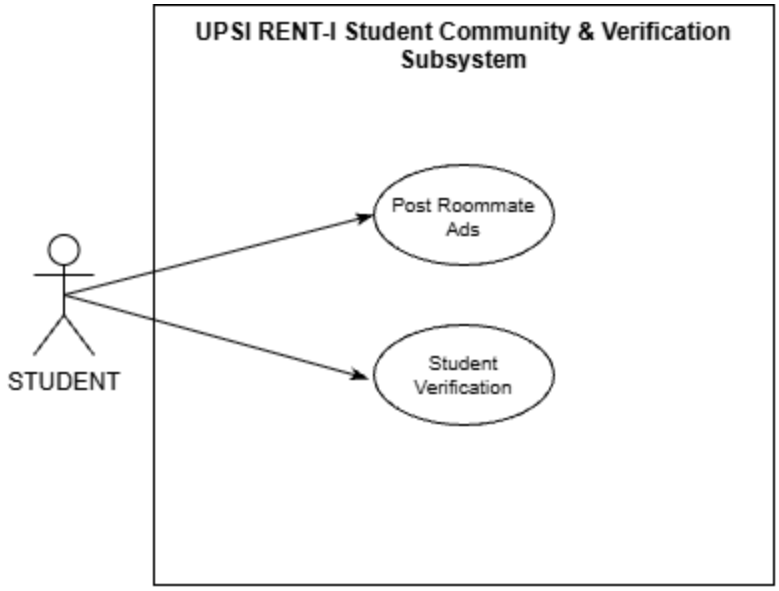


Figure 3.4.1.4 Student Community & Verification Subsystem

3.4.1.5 Admin Moderation Subsystem

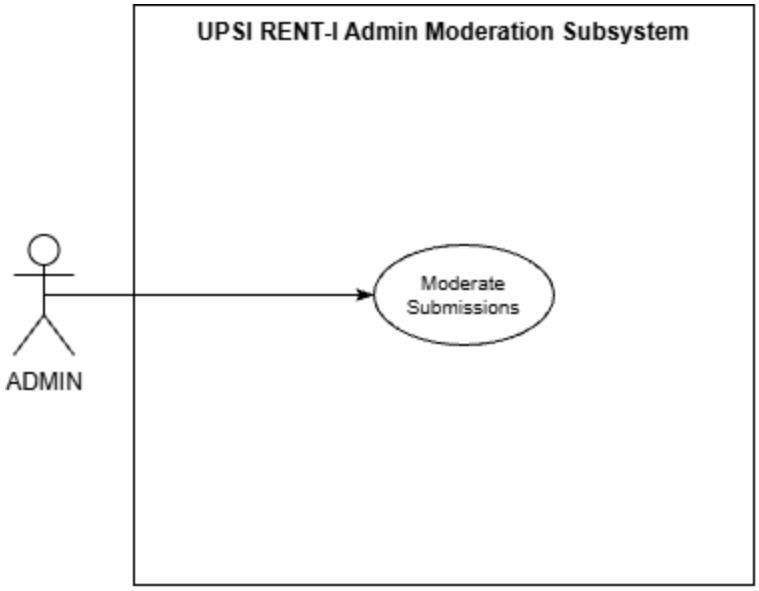
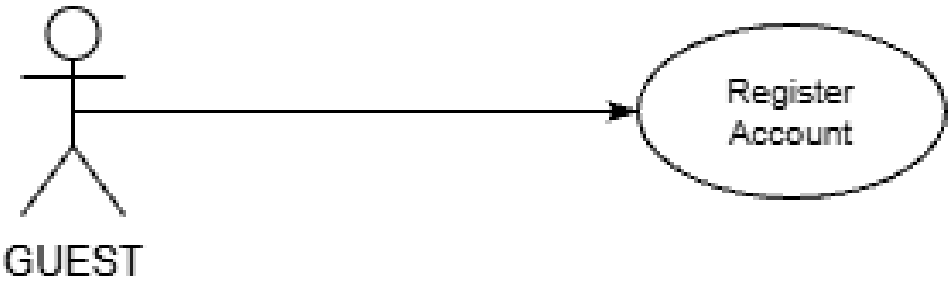


Figure 3.4.1.5 Admin Moderation Subsystem

3.4.2 Use Case Descriptions

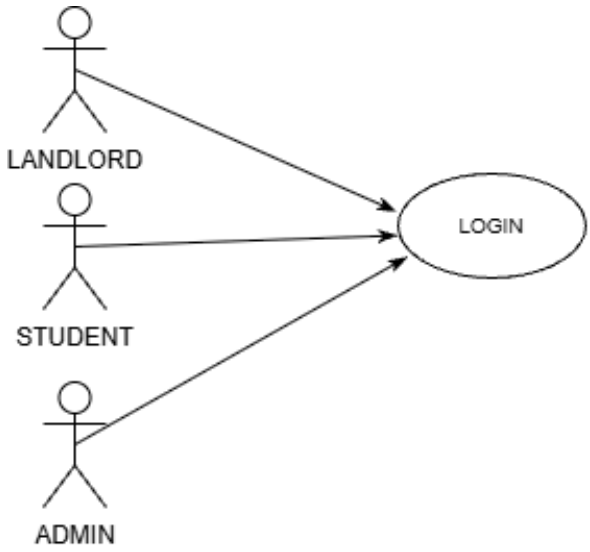
3.4.2.1 Use Case Description: UC001 - Register Account



Use Case ID	UC001	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Register Account	**Written Date**	18/10/2026
Actor	Guest (Student or Landlord)		
Description	This use case describes how a new user creates an account on the UpsiRent-I platform as either a Student or a Landlord.		
Pre-Condition	The user is not logged in and has navigated to the registration page.		
Steps	**Basic Flow:** 1. User selects "Register" from the navigation bar. 2. System displays the account registration form. 3. User selects account role (Student or Landlord). 4. User enters full name, email address, phone number, and password. 5. User reviews and checks the PDPA compliance consent checkbox. 6. User clicks "Submit Registration". 7. System validates input data and checks for existing email		

	accounts. 8. System hashes the password and saves the new user profile in the database. 9. System displays a registration success message and redirects to the Login page. **Alternate / Exception Flow:** 7a. Email already exists: System displays error. 5a. PDPA consent unticked: System prevents form submission and displays warning.
Post-Condition	A new user account record is created in the database with the designated role.
Exception	Invalid input formatting: System highlights missing/malformed fields.
Business Rules	1. Admin accounts cannot be created via public registration. 2. Passwords must be encrypted before database storage. 3. PDPA consent is mandatory for account creation.

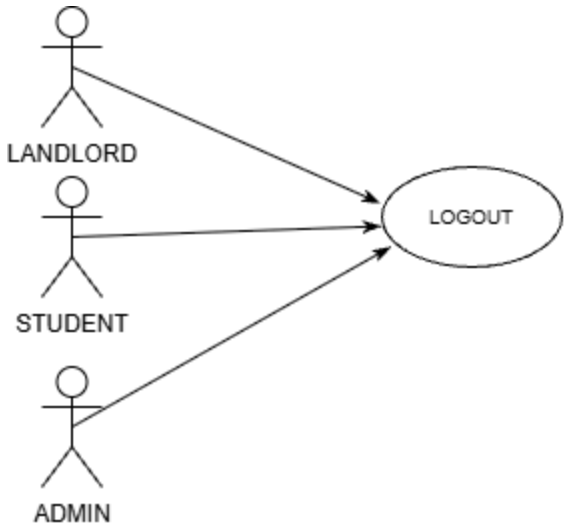
3.4.2.2 Use Case Description: UC002 - Login



Use Case ID	UC002	**Author**	Muhammad Aiman Hafiz Bin Rosli

Use Case Name	Login	**Written Date**	18/10/2026
Actor	Registered Student, Landlord, Admin		
Description	This use case describes how registered users authenticate into the UpsiRent-I system to gain role-based access to features.		
Pre-Condition	The user has an existing active account in the system database.		
Steps	**Basic Flow:** 1. User clicks the "Login" button. 2. System displays the login modal/page. 3. User inputs registered email address and password. 4. User clicks "Login". 5. System verifies credentials against the database. 6. System retrieves user role and creates an authenticated session token. 7. System redirects user to their respective dashboard and reveals role-specific UI options. **Alternate / Exception Flow:** 5a. Incorrect email or password: System displays error. 5b. Account suspended: System displays error.		
Post-Condition	User is authenticated into the session with access granted according to RBAC.		
Exception	Authentication service connection failure.		
Business Rules	1. Sessions expire after inactive timeout. 2. Passwords are compared using secure hash validation.		

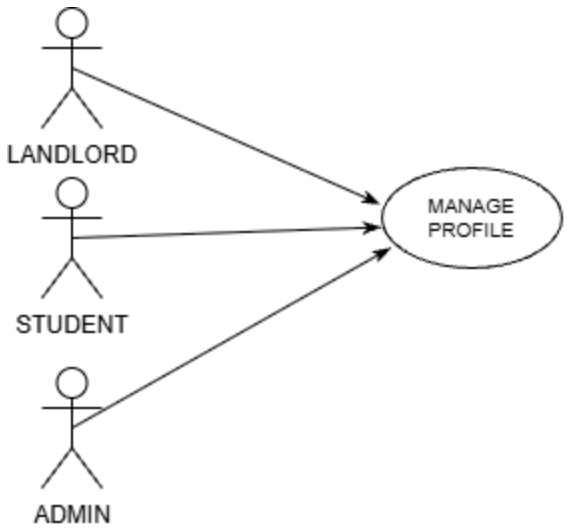
3.4.2.3 Use Case Description: UC003 - Logout



Use Case ID	UC003	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Logout	**Written Date**	18/10/2026
Actor	Registered Student, Landlord, Admin		
Description	This use case describes how an authenticated user safely terminates their session on the platform.		
Pre-Condition	The user is currently logged into an active session.		
Steps	**Basic Flow:** 1. User clicks on their profile menu or user navigation bar. 2. User selects "Logout". 3. System destroys the user's session token and clears client-side authorization state. 4. System redirects the user to the public homepage as a Guest. **Alternate / Exception Flow:** 1a. Session already expired: System redirects to login page automatically.		
Post-Condition	User session is invalidated and access is revoked until next login.		
Exception	Unexpected client-side session state error during logout execution.		

Business Rules	All session data stored in local memory or cookies must be completely cleared.

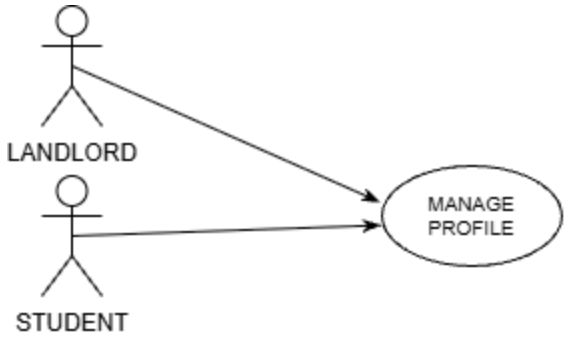
3.4.2.4 Use Case Description: UC004 - Manage Profile



Use Case ID	UC004	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Manage Profile	**Written Date**	18/10/2026
Actor	Registered Student, Landlord, Admin		
Description	This use case describes how logged-in users view and update their personal account details, phone number, and preferences.		
Pre-Condition	The user is authenticated and logged into the system.		
Steps	**Basic Flow:** 1. User navigates to "Profile Settings". 2. System displays current account information. 3. User modifies editable fields (e.g. phone number, full name). 4. User clicks "Save Changes". 5. System validates updated data and updates database record. 6. System displays success notification. **Alternate / Exception Flow:**		

	5a. Invalid phone format: System displays error.
Post-Condition	User profile information is updated in database.
Exception	Database update error during profile details save.
Business Rules	Primary email address cannot be changed directly without re-verification.

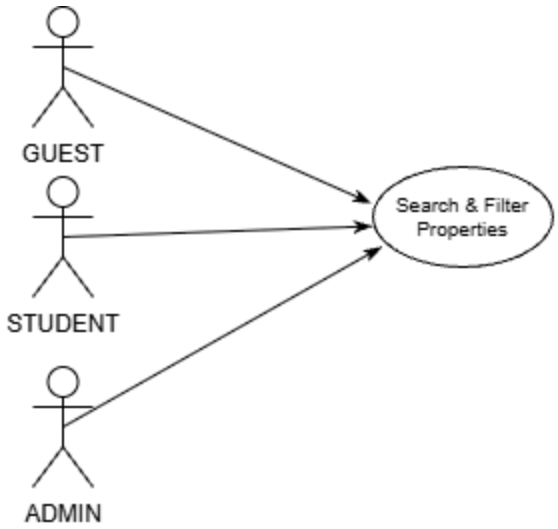
3.4.2.5 Use Case Description: UC005 - Reset Password



Use Case ID	UC005	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Reset Password	**Written Date**	18/10/2026
Actor	Registered Student, Landlord		
Description	This use case describes how a user who forgot their password can reset it by requesting a verification link/code.		
Pre-Condition	The user has access to the email address associated with their account.		
Steps	**Basic Flow:** 1. User clicks "Forgot Password?". 2. System prompts user to enter their registered email address. 3. User submits email address. 4. System generates a password reset token and sends a reset		

	email. 5. User clicks the reset link in their email. 6. System presents the new password entry form. 7. User enters and confirms the new password. 8. System hashes the new password, updates the database, and invalidates the reset token. 9. System displays confirmation and prompts user to log in. **Alternate / Exception Flow:** 2a. Email not found: System displays error. 5a. Expired reset token: System displays error.
Post-Condition	User password is updated securely in the database.
Exception	Mail server fails to deliver the reset email.
Business Rules	1. Reset tokens expire after 24 hours. 2. Passwords must be hashed prior to storage.

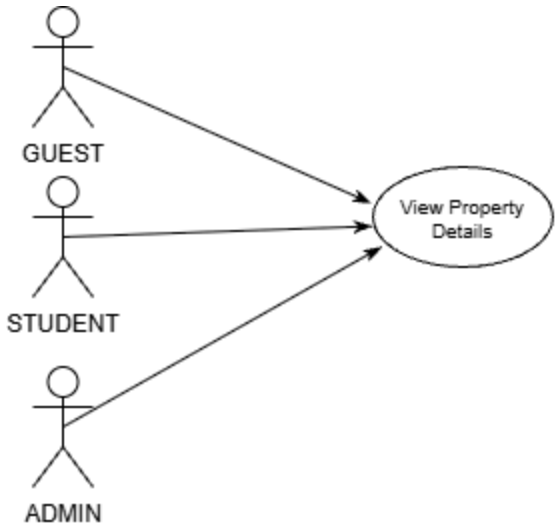
3.4.2.6 Use Case Description: Search & Filter Properties



Use Case ID	UC006	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Search & Filter Properties	**Written Date**	18/10/2026

Actor	Guest, Registered Student, Verified Student, Admin
Description	Allows users to browse, search, and filter property listings (Rumah Sewa, Homestay, Cari Rakan Bilik) based on specific criteria.
Pre-Condition	User is on the main property listing page.
Steps	**Basic Flow:** 1. User selects a category tab. 2. System loads the corresponding approved listings from the database. 3. User applies filters (Location, Availability, Budget, or Keywords). 4. System dynamically updates the property grid to display only matching results. **Alternate / Exception Flow:** 3a. No matching results: 1. System displays a "No Properties Found" message. 2. User clicks the "Reset" button to clear all filters.
Post-Condition	Filtered property listings are displayed to the user.
Exception	Database connection timeout during filtering query.
Business Rules	1. Only properties with "Approved" status by Admin are visible. 2. Verified Students will see prices with a 10% discount automatically applied.

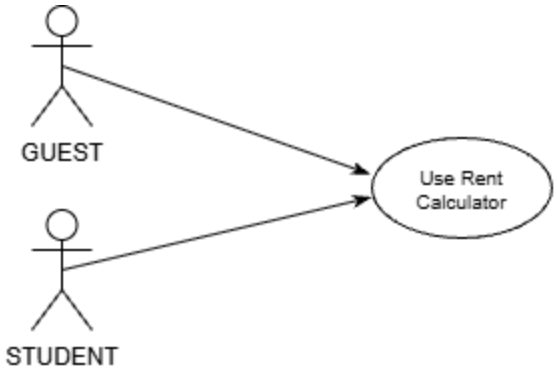
3.4.2.7 Use Case Description: View Property Details



Use Case ID	UC007	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	View Property Details	**Written Date**	18/10/2026
Actor	Guest, Registered Student, Verified Student, Admin		
Description	Displays comprehensive information about a selected property, including photo galleries, amenities, map links, and landlord contact details.		
Pre-Condition	User is browsing the property listing page.		
Steps	**Basic Flow:** 1. User clicks on a specific property card. 2. System opens a detailed interactive modal or dedicated page for the property. 3. User views the photo gallery, inventory checklist, and tenant reviews. 4. User clicks the "View on Map" button. 5. System redirects the user to the Google Maps URL scheme using the property's location parameters. **Alternate / Exception Flow:** None.		

Post-Condition	Detailed property information is fully displayed.
Exception	Failed to load property images due to cloud storage unavailability.
Business Rules	If the user is a Guest, certain contact details may require logging in or acknowledging PDPA consent before being fully revealed.

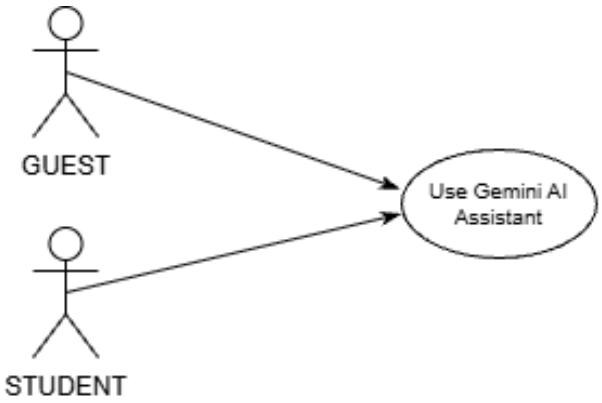
3.4.2.8 Use Case Description: Use Rent Calculator



Use Case ID	UC008	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Use Rent Calculator	**Written Date**	18/10/2026
Actor	Guest, Registered Student, Verified Student		
Description	Allows users to estimate the per-head rental cost by adjusting sliders and toggling shared utilities.		
Pre-Condition	User is viewing a specific Property Details page.		
Steps	**Basic Flow:** 1. User interacts with the Rent Calculator widget. 2. User adjusts the base rent slider and selects the number of tenants. 3. User ticks checkboxes for shared utilities. 4. System dynamically recalculates the total per-head cost. 5. User clicks "Share via WhatsApp".		

	6. System generates a pre-filled text template and opens the WhatsApp URL scheme. **Alternate / Exception Flow:** None.
Post-Condition	Accurate rental estimates are generated and ready for external sharing.
Exception	Device lacks a compatible WhatsApp application when sharing is triggered.
Business Rules	Calculator is an optional extension (<<extend>>) of the View Property Details use case.

3.4.2.9 Use Case Description: Use Gemini AI Assistant



Use Case ID	UC009	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Use Gemini AI Assistant	**Written Date**	18/10/2026
Actor	Guest, Registered Student, Verified Student		
Description	Provides an interactive AI chat interface to analyze rental contracts for red flags and offer personalized renting advice.		

Pre-Condition	User is viewing a specific Property Details page.
Steps	**Basic Flow:** 1. User clicks the Gemini AI Assistant floating action button. 2. System opens the chat interface overlay. 3. User types a query or uploads a rental contract snippet. 4. System backend injects the current property context and forwards the prompt to the Google Gemini API proxy. 5. System receives the AI response and displays it in the chat interface. **Alternate / Exception Flow:** 4a. Gemini API Unreachable: 1. System catches the timeout/error. 2. System falls back to predefined local rule-based advice or displays "AI Assistant is currently unavailable".
Post-Condition	Context-aware AI response is generated and displayed to the user.
Exception	Rate limiting or token exhaustion from the Google Gemini API.
Business Rules	1. API keys must remain hidden on the server-side. 2. AI Assistant is an optional extension (<<extend>>) of the View Property Details use case.

3.4.2.10 Use Case Description: Submit Property Listing



Use Case ID	UC010	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Submit Property Listing	**Written Date**	18/10/2026

Actor	Landlord
Description	Allows verified property owners/agents to submit new accommodation advertisements.
Pre-Condition	User is logged in as a Landlord.
Steps	**Basic Flow:** 1. Landlord navigates to the "Submit Listing" dashboard. 2. Landlord fills in property details (Name, Address, Price, Capacity, Facilities). 3. Landlord uploads multi-image property photos. 4. Landlord ticks the mandatory "Accept PDPA Consent" checkbox (<<include>>). 5. Landlord clicks "Submit". 6. System validates the inputs and saves the record in the database with a "Pending" status. 7. System displays a success notification. **Alternate / Exception Flow:** 4a. PDPA Consent Unticked: 1. System blocks submission and displays a warning prompt.
Post-Condition	Property listing is saved in the database awaiting Admin moderation.
Exception	Image upload fails due to exceeding file size limits or unsupported formats.
Business Rules	1. Listings will not be visible to the public until manually "Approved" by an Admin. 2. PDPA explicit consent is mandatory before submission.

3.4.2.11 Use Case Description: Post Roommate Ads



Use Case ID	UC011	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Post Roommate Ads	**Written Date**	18/10/2026
Actor	Registered Student, Verified Student		
Description	Allows students to publish advertisements looking for roommates to share an existing rental.		
Pre-Condition	User is logged in with a Student role.		
Steps	**Basic Flow:** 1. Student navigates to the "Roommate Hub" and clicks "Create Ad". 2. Student enters requirements (Preferences, Budget, Move-in Date, Contact Info). 3. Student ticks the mandatory "Accept PDPA Consent" checkbox (<<include>>). 4. Student clicks "Post". 5. System saves the advertisement in the database with a "Pending" status. 6. System displays a success notification. **Alternate / Exception Flow:** 3a. PDPA Consent Unticked: 1. System prevents form submission and displays an error message.		
Post-Condition	Roommate advertisement is saved in the database awaiting Admin moderation.		
Exception	Database writing failure.		
Business Rules	1. Ads must be moderated by an Admin before public display to ensure community safety. 2. PDPA consent is required to display personal contact information.		

3.4.2.12 Use Case Description: Student Verification



Use Case ID	UC012	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Student Verification	**Written Date**	18/10/2026
Actor	Registered Student		
Description	Allows students to upload proof of enrollment for Admin verification to unlock a 10% rental discount.		
Pre-Condition	User is logged in as a Registered Student.		
Steps	**Basic Flow:** 1. Student navigates to the "Verification" section. 2. Student uploads a clear image of their Student ID or Offer Letter. 3. Student ticks the mandatory "Accept PDPA Consent" checkbox (<<include>>). 4. Student clicks "Submit for Verification". 5. System updates the user's verification status to "Pending" and securely stores the document. 6. System notifies the student that the submission is under review. **Alternate / Exception Flow:** 2a. File type/size error: 1. System blocks upload and specifies valid formats (JPEG/PNG/PDF).		
Post-Condition	Student's verification status is set to Pending and the document is ready for Admin review.		

Exception	Cloud storage failure preventing document upload.
Business Rules	1. Verification documents are strictly handled according to PDPA guidelines and will be auto-deleted after a retention period. 2. Only verified students are eligible for the system-wide 10% discount.

3.4.2.13 Use Case Description: Moderate Submissions



Use Case ID	UC013	**Author**	Muhammad Aiman Hafiz Bin Rosli
Use Case Name	Moderate Submissions	**Written Date**	18/10/2026
Actor	Admin		
Description	Allows Administrators to review and approve/reject pending student verifications, property listings, and roommate ads.		
Pre-Condition	Admin is logged into the Admin Dashboard.		
Steps	**Basic Flow:** 1. Admin navigates to the "Moderation Dashboard". 2. System displays lists of items with "Pending" status (Verifications, Properties, Roommate Ads). 3. Admin selects an item to review its details and attached documents/photos. 4. Admin clicks either "Approve" or "Reject" (with an optional reason). 5. System updates the item's status in the database accordingly. 6. System triggers an in-app notification to the respective user		

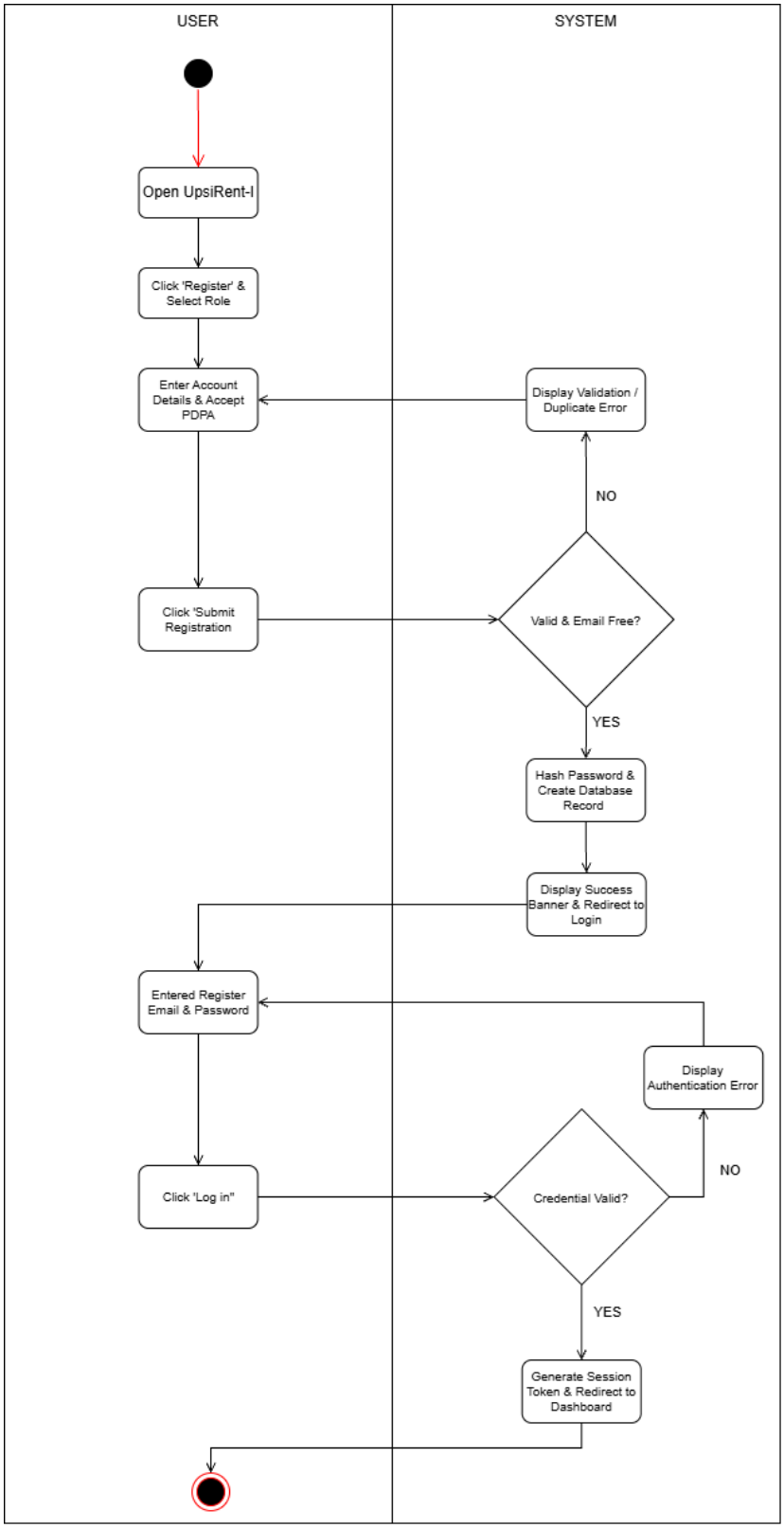
	regarding the status change. **Alternate / Exception Flow:** None.
Post-Condition	The status of the submitted item is finalized (Approved or Rejected) and the user is notified.
Exception	Database transaction deadlock during status update.
Business Rules	1. Approved properties and roommate ads immediately become visible to the public. 2. Approved student verifications automatically upgrade the user role to "Verified Student" and unlock discounts.

3.5 Data and Behavioral Models

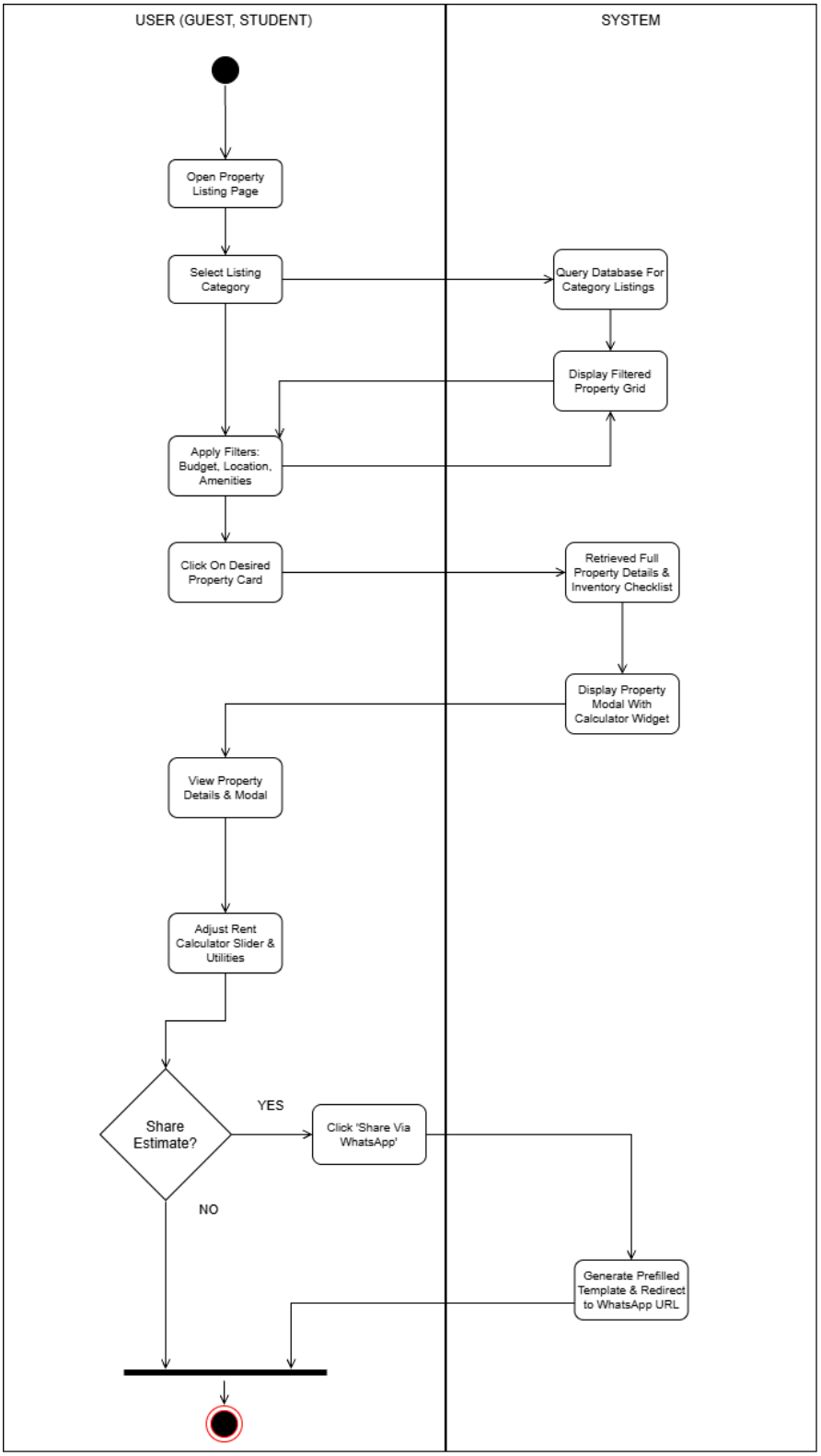
This section presents the structural, behavioral, and data persistence models of the UniWise system. The system behavior is specified using Activity Diagrams with swimlane partitions, while the system structure and database schema are represented via a Class Diagram and a Logical Database Table.

3.5.1 Activity Diagram

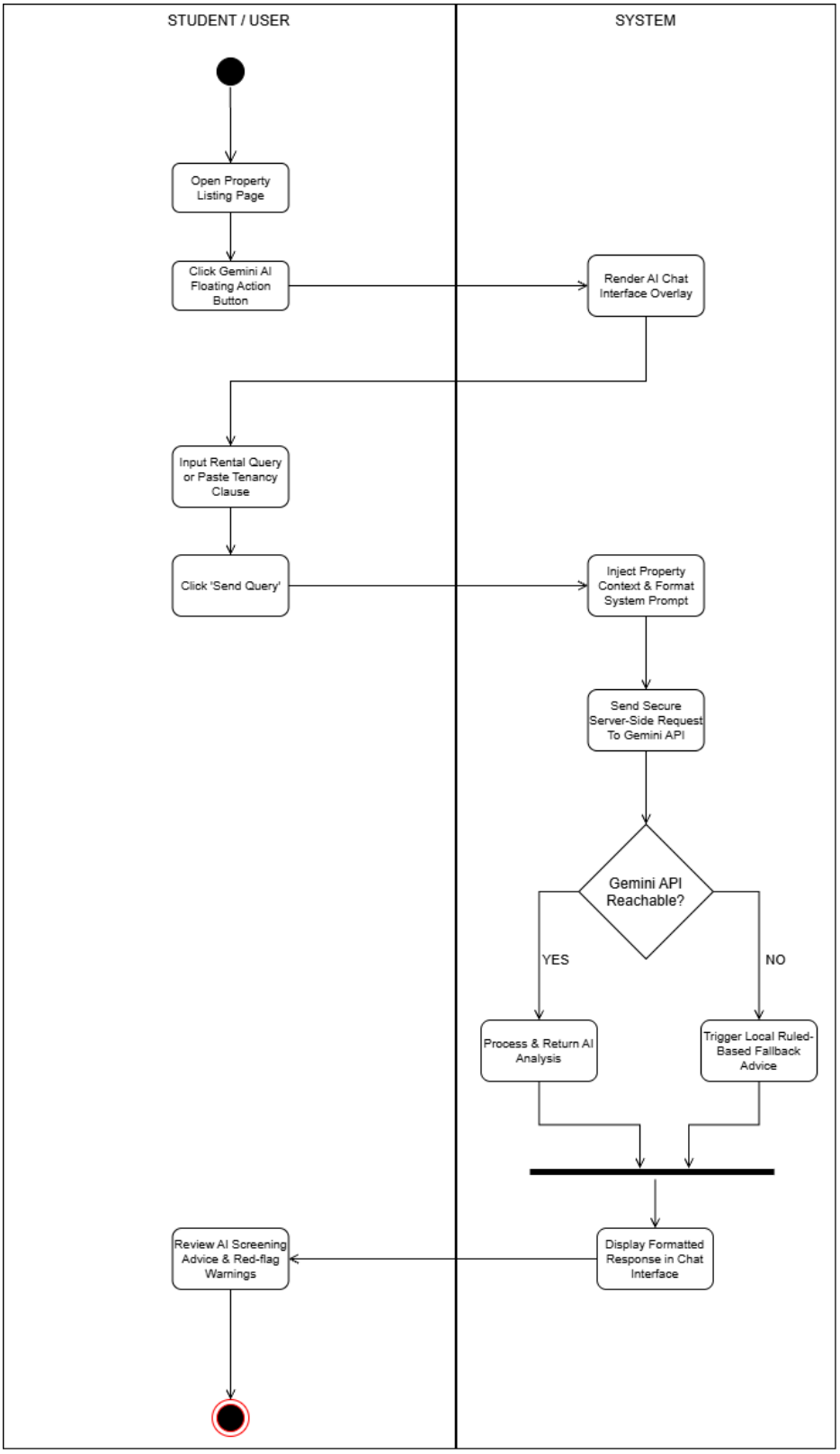
3.5.1.1 Login



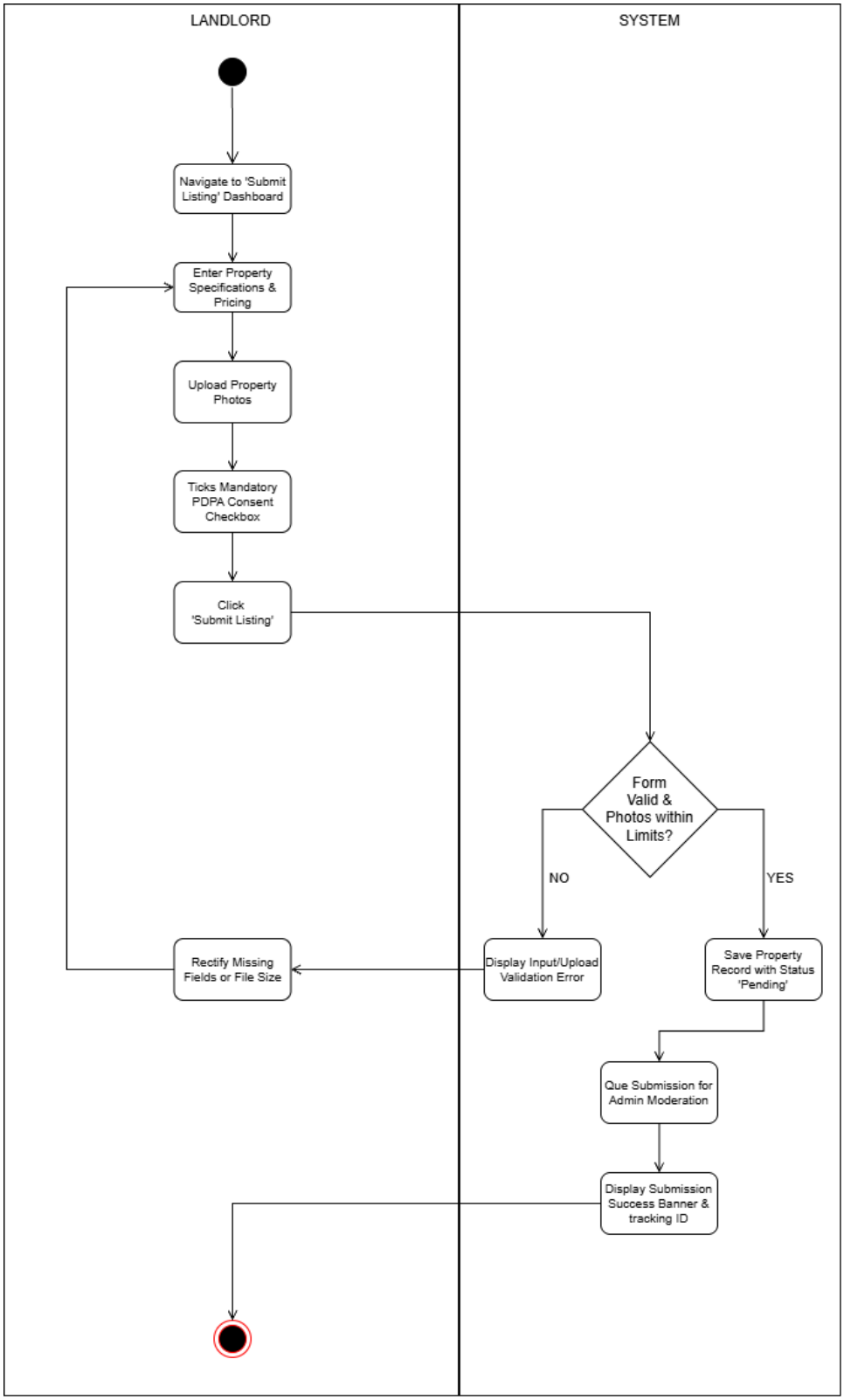
3.5.1.2 Property Search & Rent Calculator



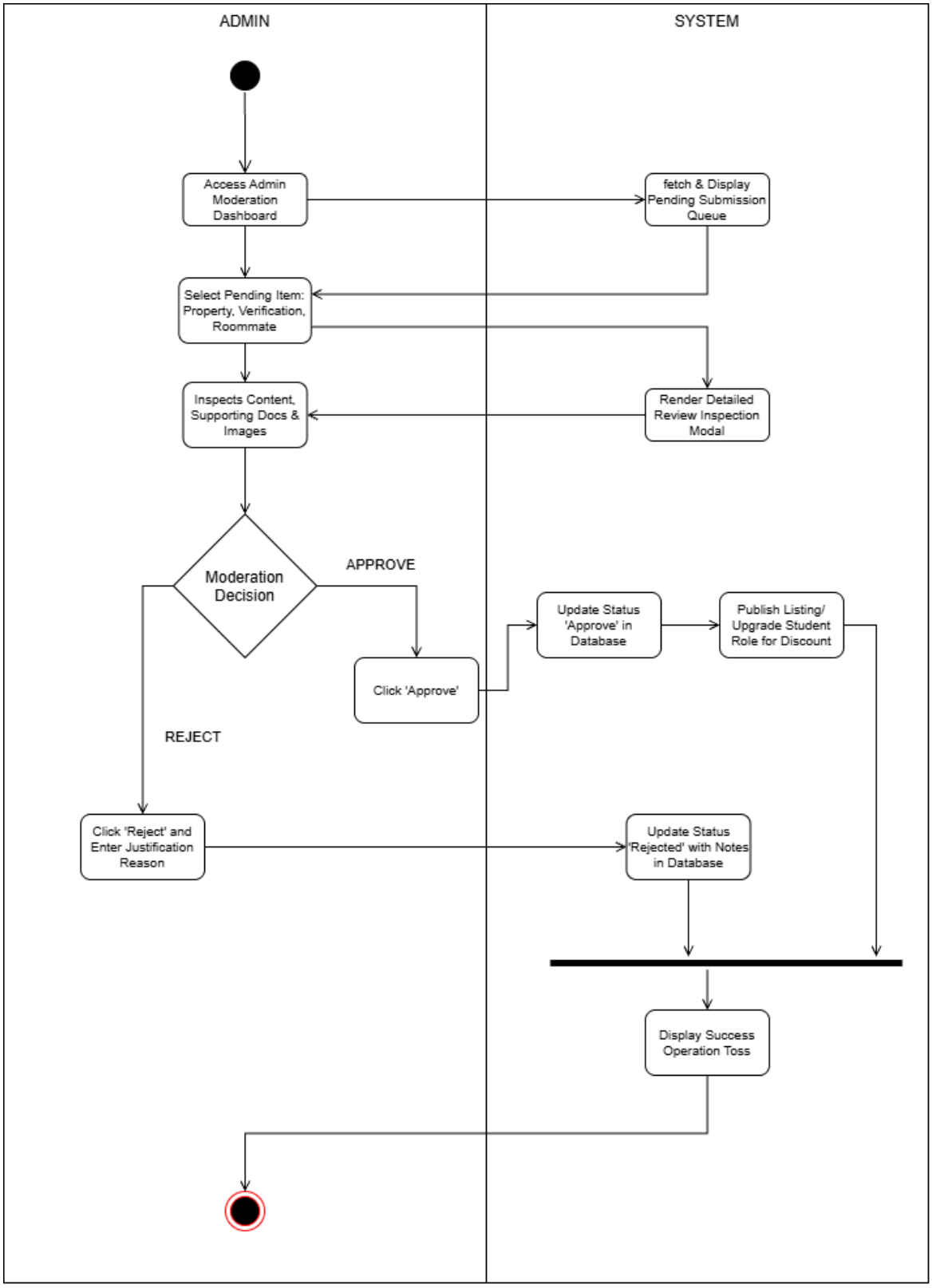
3.5.1.3 Gemini AI Assistant Screening



3.5.1.4 Submit Property Listing (Landlord)



3.5.1.5 Admin Moderation (Approve/Reject)


ADMIN
SYSTEM
Access Admin Moderation Dashboard
fetch & Display Pending Submission Queue
Select Pending Item: Property, Verification, Roommate
Inspects Content, Supporting Docs & Images
Render Detailed Review Inspection Modal
Moderation Decision
APPROVE
REJECT
Click 'Approve'
Update Status 'Approve' in Database
Publish Listing/ Upgrade Student Role for Discount
Click 'Reject' and Enter Justification Reason
Update Status 'Rejected' with Notes in Database
Display Success Operation Toss

3.5.2 Class Diagram

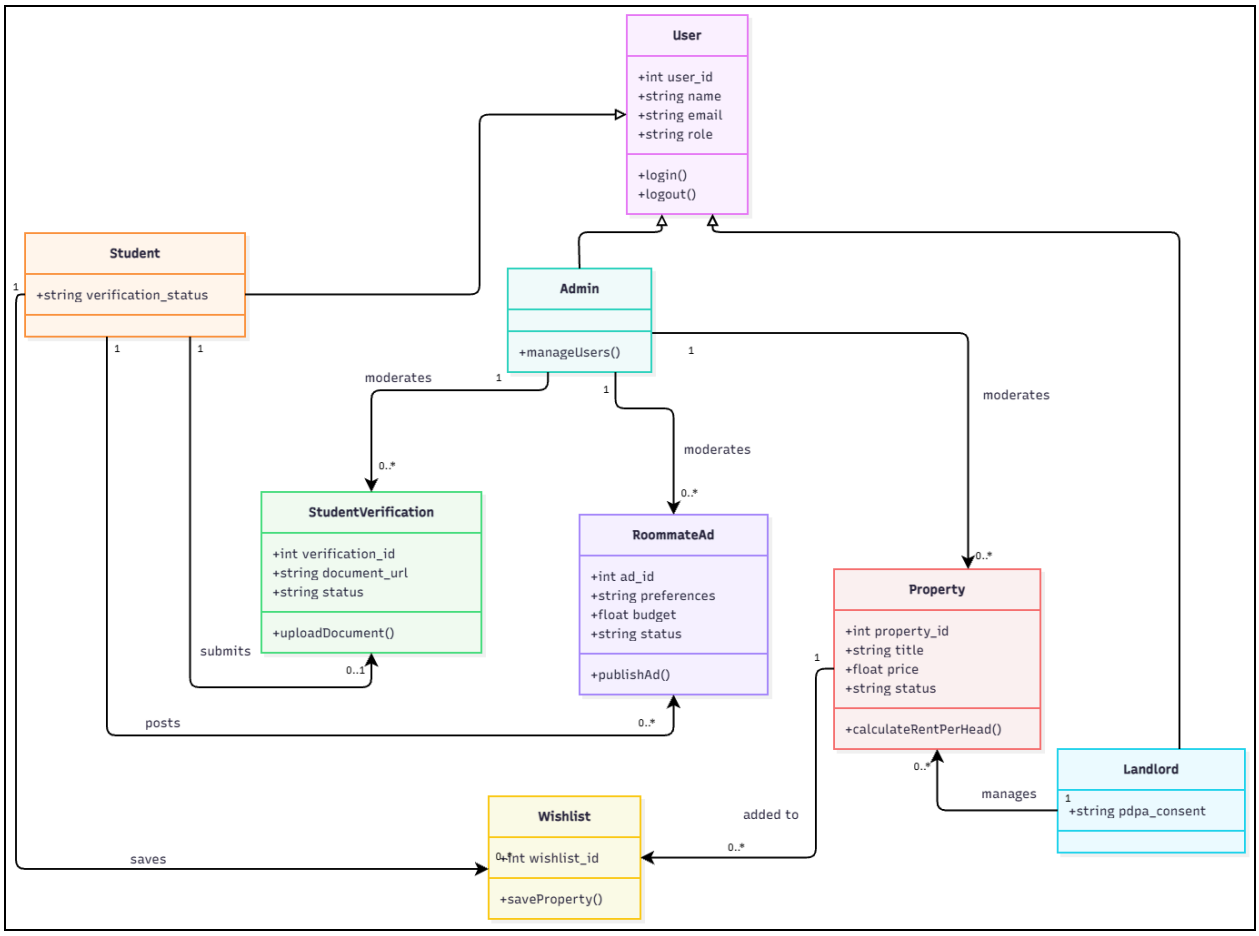


Figure 3.5.2: UpsiRent-I Class Diagram

The structural layout of the UpsiRent-I system is detailed in the Class Diagram shown in Figure 3.5.2. Built upon the fundamental concept of inheritance, the parent class `User` generalizes three main system roles: `Student`, `Landlord`, and `Admin`. Furthermore, the diagram outlines the operational data relationships connecting these user roles with essential domain models, including `Property`, `RoommateAd`, `StudentVerification`, and `Wishlist`.

3.5.3 Logical Database

This section outlines the logical database schema for the UpsiRent-I system. It defines the core tables, primary keys (PK), foreign keys (FK), and a brief description of each entity.

Table Name	Key Fields	Description
Users	user_id (PK)	Stores account credentials, personal details, and RBAC roles (Student, Landlord, Admin).
Properties	property_id (PK) landlord_id (FK -> Users)	Stores accommodation listings including pricing, location, capacity, and approval status.
RoommateAds	ad_id (PK) student_id (FK -> Users)	Stores student advertisements looking for roommates, including budget and preferences.
StudentVerifications	verification_id (PK) student_id (FK -> Users)	Stores uploaded student IDs or offer letters for manual admin verification to unlock the 10% discount.
Wishlists	wishlist_id (PK) student_id (FK -> Users) property_id (FK -> Properties)	Junction table linking students to their saved property listings for future reference.

Table 3.5.3: Database Schema Definitions

3.6 Design Constraint Requirements

This section outlines the technological limitations, architectural boundaries, and compliance constraints that shape the development of the UpsiRent-I system.

Constraint ID	Requirement
DC001	The application must be developed entirely as a no-code solution using Google AppSheet, ensuring mobile-responsive compatibility across all devices without hand-written UI code.
DC002	The system's data model must utilize Google Sheets as the primary structured database, serving as the backend inventory and user management ledger.
DC003	User authentication must strictly utilize Google Workspace for Education Plus accounts via AppSheet's built-in secure sign-in requirements.
DC004	All API calls to Google Gemini must be executed server-side to prevent the exposure of sensitive API keys to the client browser.
DC005	The collection and retention of personal data (e.g., Student IDs, contact numbers) must comply with the Personal Data Protection Act 2010 (PDPA), requiring explicit user consent via checkboxes prior to data submission.

Table 3.6 Design Constraint Requirements

3.7 Quality Requirements

This section defines the non-functional criteria that ensure the system is secure, reliable, performant, and user-friendly.

Quality Attribute	Requirement
Performance	The system shall return search and filter results within 2 seconds. AI chat responses from the Gemini API must be displayed within 5 seconds under normal network conditions.
Security	All user passwords must be securely hashed using bcrypt prior to database storage. Data transmission must be encrypted using HTTPS/SSL protocols.
Reliability	The system shall maintain an uptime availability of 99% during typical academic semesters to ensure students have continuous access to housing resources.
Usability	The interface shall incorporate clear navigation, modal dialogs for details, and large text accessibility modes to ensure a seamless user experience without requiring a steep learning curve.
Maintainability	The codebase shall be structured following the MVC (Model-View-Controller) architecture to allow for easy updates, scalability, and future feature integrations.

Table 3.7 Quality Requirements